

Learning Outcomes/OBJECTIVES

- ❖ Hemoglobin (Structure)
- ❖ O₂ & CO₂ transport
- ❖ Lung surfactants
- Definition
- Chemical Composition
- Application
- Function Clinical Utility

INTRODUCTION

❖ **Myo** - from the Greek word for "muscle"

❖ **Hemo** - from the Greek word for "blood"

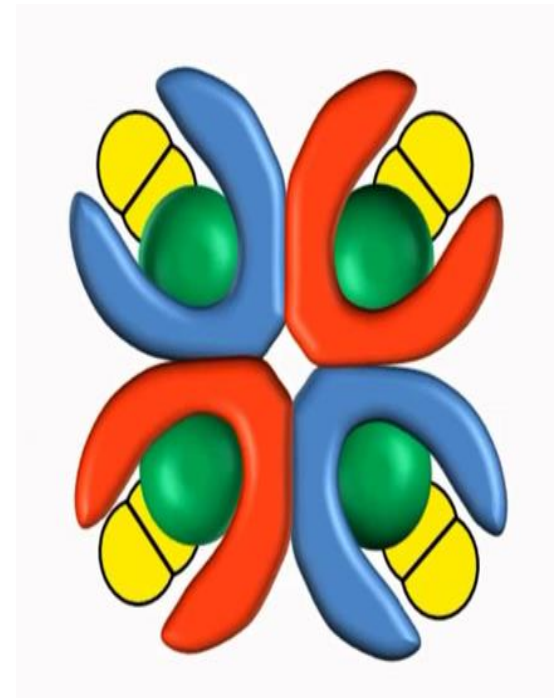
- Myoglobin and hemoglobin are hemoproteins (globular proteins) whose physiological importance is principally related to their ability to bind molecular oxygen.
- These globins dramatically improve the concentration of molecular oxygen that can be dissolved in the biological fluids of vertebrates and some invertebrates.

Hemoglobin & Myoglobin

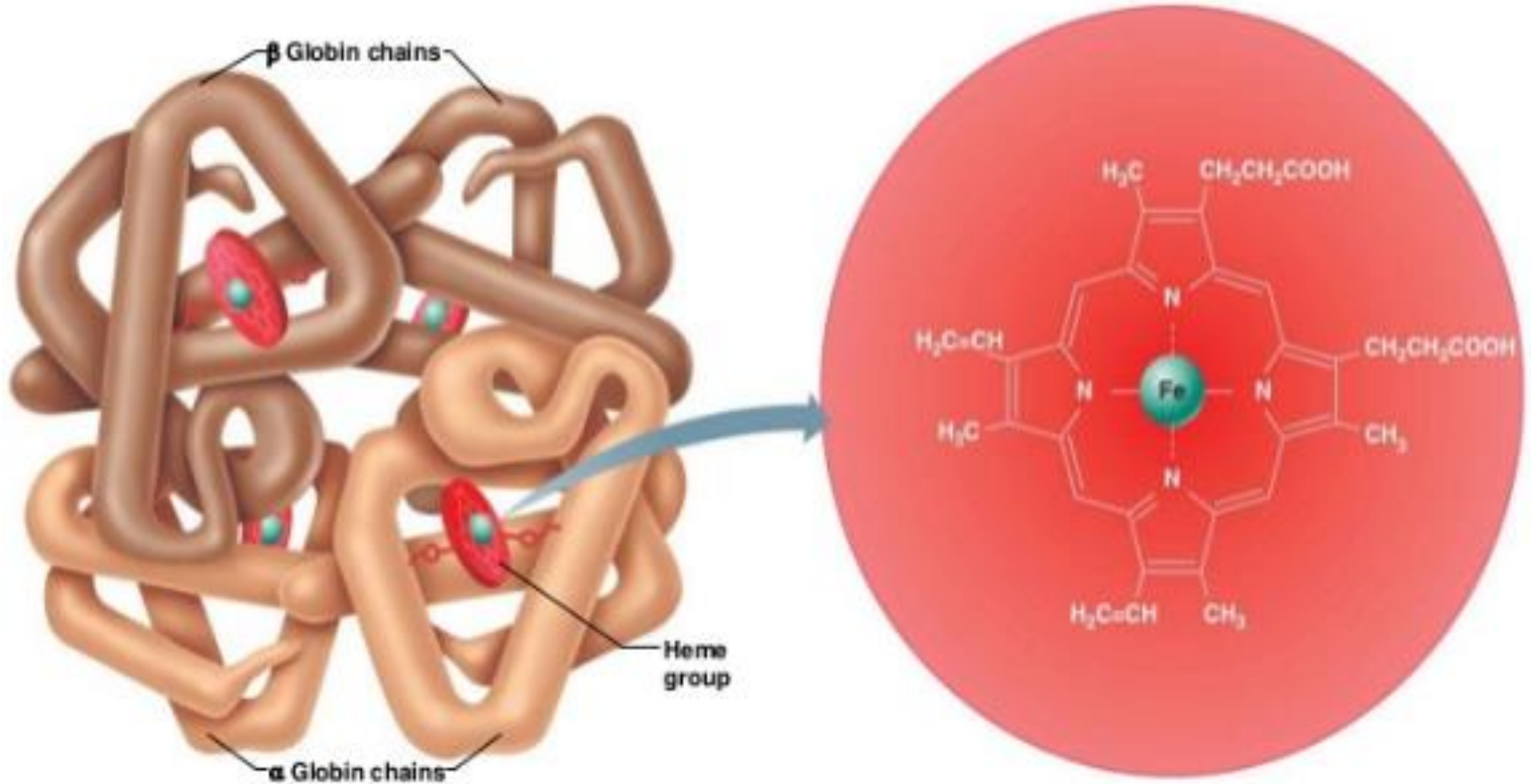
S.No	Hemoglobin (Hb)	Myoglobin (Mb)
1.	Hb is Oxygen transport protein in RBCs of blood.	Mb is Oxygen storing protein in muscles.
2.	Tetrameric has four Heme and binds with 4O₂	Monomeric has one Heme and binds with 1 O₂.
3.	Oxygenated at Lungs	Oxygenated at Muscle Cell Cytosol.
4.	HbO ₂ unloads oxygen at tissues when pO ₂ is at 40 mmHg. P₅₀ for HbA₁ is 27 torr.	MbO ₂ unloads oxygen at cell cytosol when pO ₂ is at 5 mmHg. to rapidly respiring cells P₅₀ for Mb is 2 torr.
5.	ODC is sigmoid shaped	ODC is hyperbolic shaped.
6.	Hb has 574 amino acids. Mol .wt-67,000 Daltons.	Mb has 153 amino acids. Mol wt-17,200 Daltons.

Hemoglobin

- Hemoglobin is red blood pigment, found **exclusively** in erythrocytes
- It is a conjugated protein (chromoprotein) containing heme as the prosthetic group & globin as the protein part- apoprotein
- Hemoglobin = Heme + globin (protein)
- The basic protein “**globin**” varies from species to species in its amino acid composition & sequence, and is responsible for species-specificity
- Polypeptide chains of globin of adult Hb, contain high content of ‘**histidine**’ & ‘**lysine**’ & small amount of **isoleucine**
- Heme containing proteins are characteristic of aerobic organism
- The polypeptide chains are usually tetrameric 2 α & 2 β chains.
- Each gram of Hb contains 3.4 mg of iron.
- ❖ **Normal levels:**
 - ✓ Adult male: 14 to 16 gm%
 - ✓ Female: 13 to 15 gm%



Structure of Hemoglobin

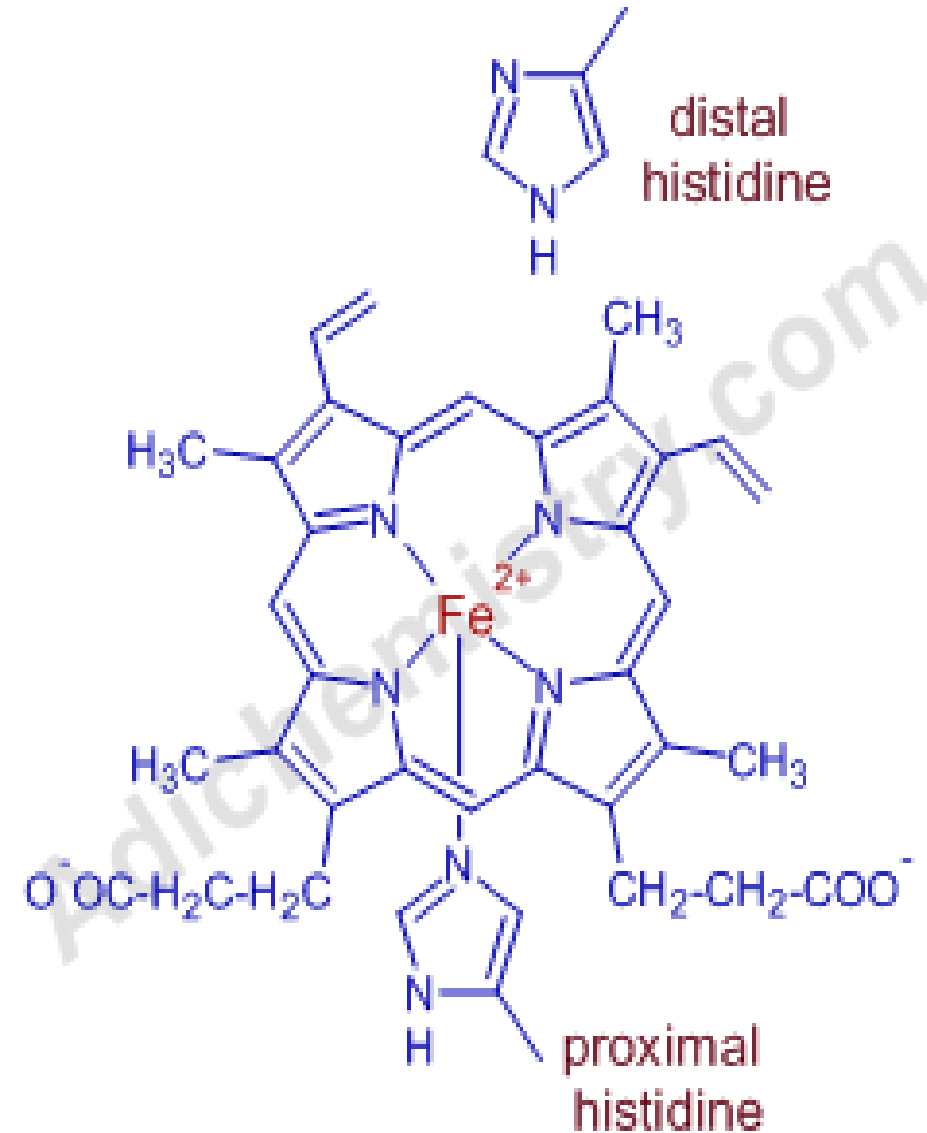


Hemoglobin

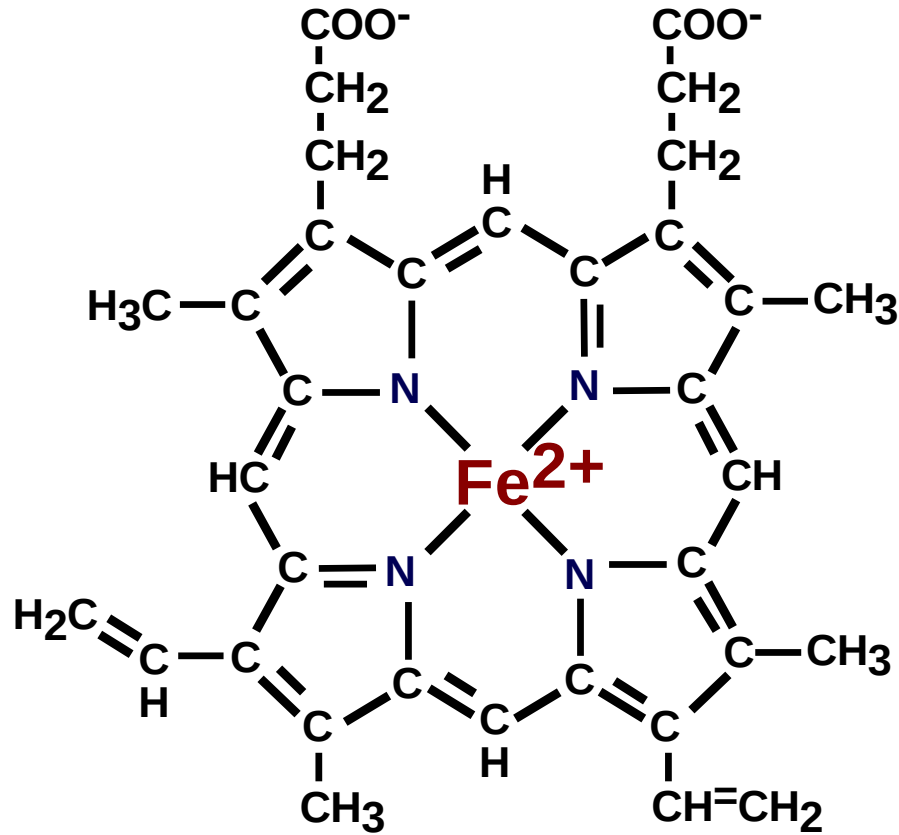
- α chain gene is on chromosome 16, beta, gamma and delta chains on chromosome 11.

Each alpha chain has 141 amino acids.

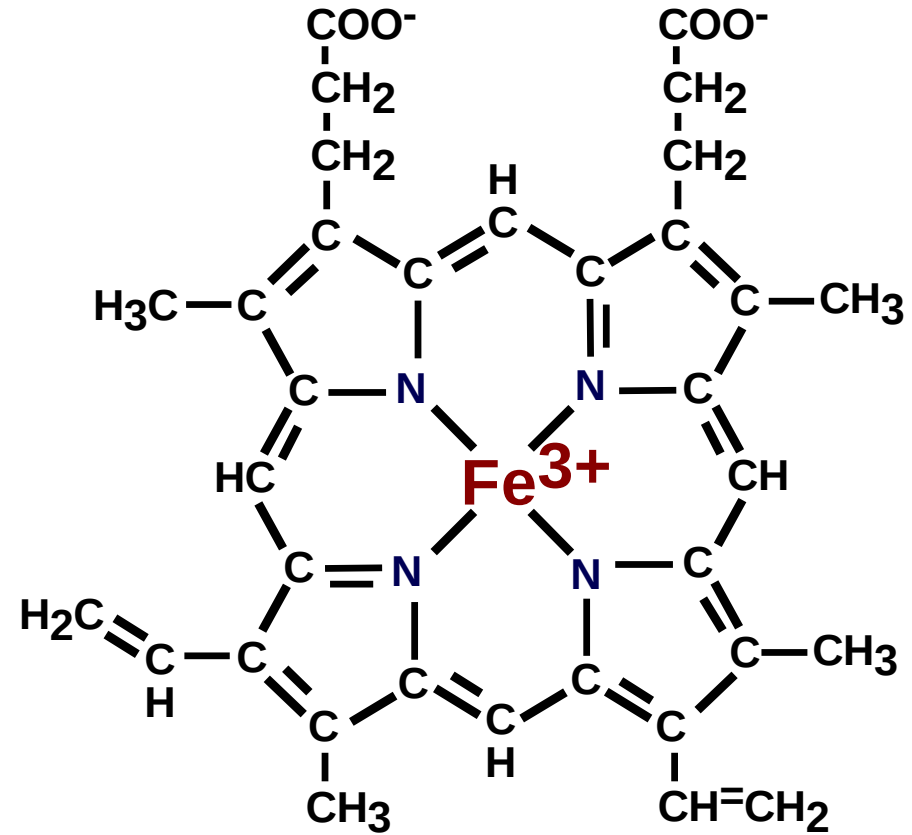
- The beta, gamma and delta chains have 146 amino acids.
- 36 histidine residues in Hb molecule=for buffering action
- 58th residue in α chain = **distal histidine** it is far away from the iron atom.
- 87th residue in α chain is = **proximal histidine**, it lies nearer to the iron atom



Iron has 2-oxidation states

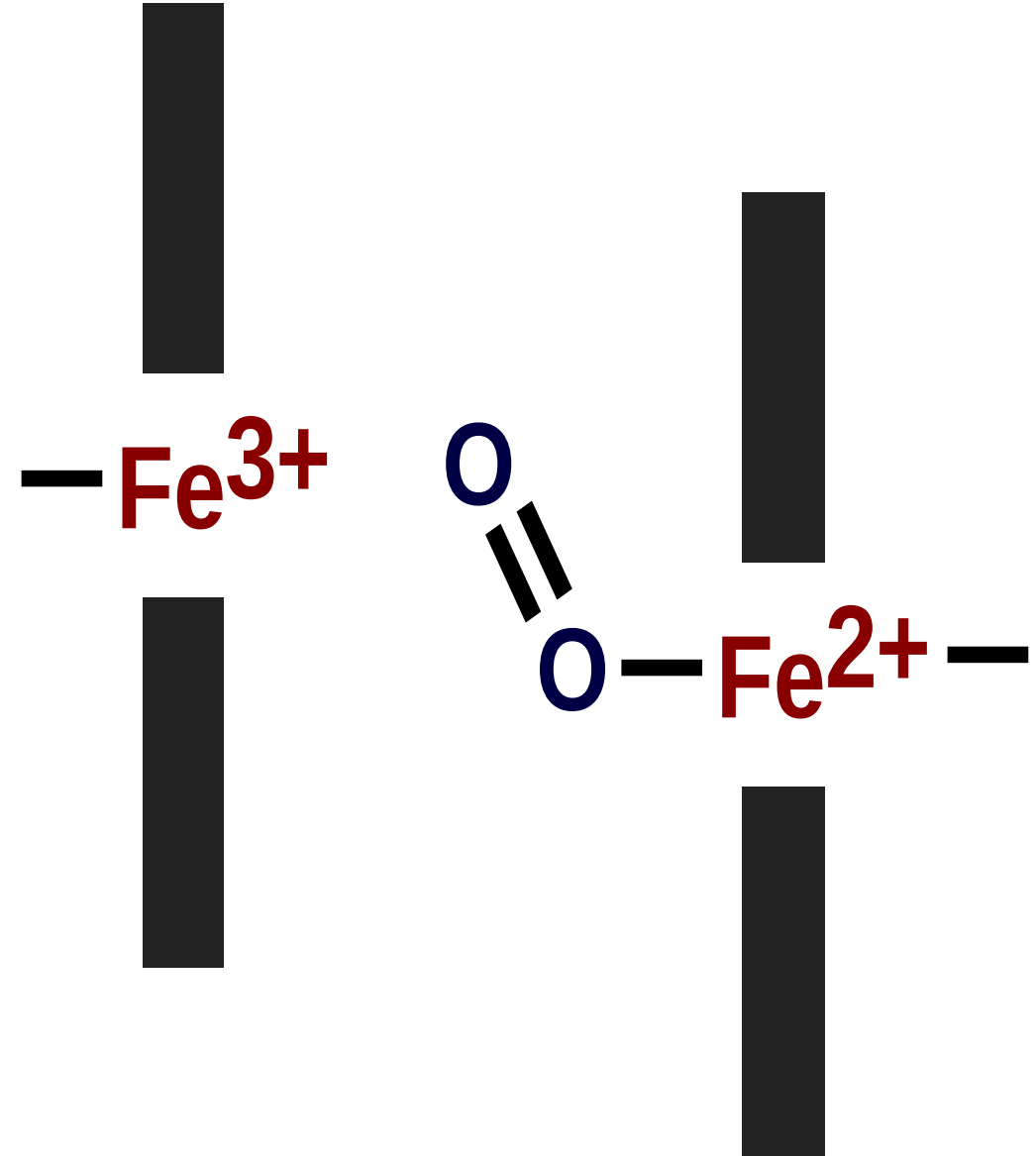


- ✓ Ferrohemoglobin (Fe²⁺)
- ✓ binds O₂



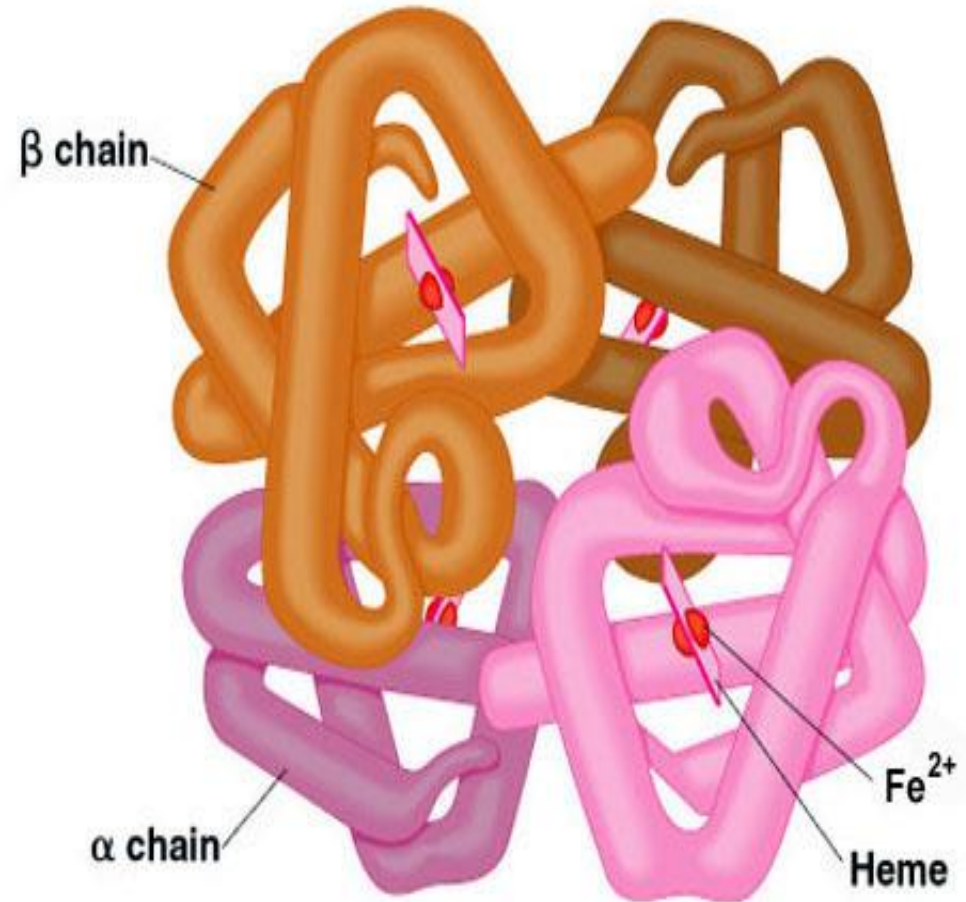
- ✓ Ferrihemoglobin (Fe³⁺), also known as methemoglobin,
- ✓ does not bind O₂

- When heme is free in solution, it can bind O_2 .
- But in the presence of a second heme molecule, O_2 will oxidize Fe^{2+} to Fe^{3+} , so the first heme no longer binds O_2
- This process continues until all of the heme iron molecules have been oxidized.
- One function of Hb & Hb is to physically separate the heme molecules, so oxygen doesn't oxidize the Fe^{2+} .



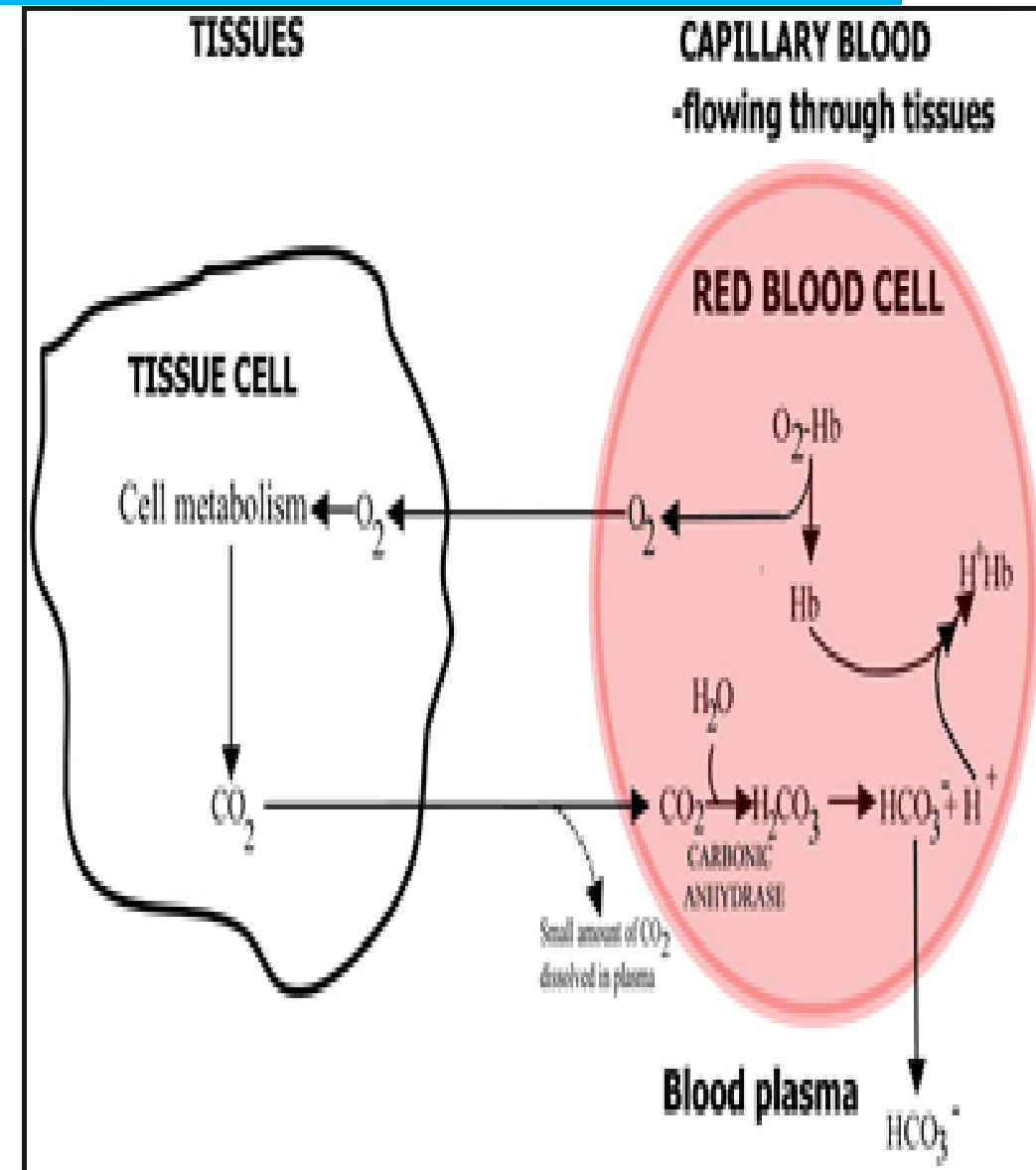
O₂ binding to Hb

- ❖ Tetrameric, 2 α -chains & 2 β -
- One molecule of Hb can bind with four molecules of O₂.
- carries O₂ from lungs to tissues
- cooperative binding of O₂
- required to increase the solubility of O₂ in blood



Transport of O₂ & CO₂ by Hemo

- Hemoglobin has all the requirements of an ideal respiratory pigment (Barcroft):
 - ✓ It can transport large quantities of O₂ from lungs to the tissues
 - ✓ Transport of CO₂ & protons from tissues to lungs for excretion
 - ✓ It has great solubility
 - ✓ It can take up and release oxygen at appropriate partial pressures
 - ✓ It is a powerful buffer.



Effect of 2,3-BPG

- Normally the 2,3-bisphosphoglycerate level is 15 ± 1.5 mg/g Hb.
- The 2,3-BPG concentration is higher in young children compared to the elderly.
- The 2,3-BPG is produced from 1,3-BPG, an intermediate of glycolytic pathway
- The 2,3-BPG, preferentially binds to deoxy-Hb and stabilizes the T conformation. When the T form reverts to the R conformation, the 2,3-BPG is ejected.
- During oxygenation, BPG is released
- The high oxygen affinity of fetal blood (HbF) is due to the inability of gamma chains to bind 2,3-BPG.

Adaptation to High Altitude

1. Increase in the number of RBCs
2. Increase in concentration of Hb inside RBCs
3. Increase in BPG

Allosteric Effects and Cooperativity

❖ **Allosteric effects** occur when the binding properties of a macromolecule change as a consequence of a second ligand binding to the macromolecule and altering its affinity towards the first, or primary, ligand.

❖ 2 Types of Allosteric effects

1. **Homotropic allosteric effect:** If the two ligands are the same (*e.g.* O₂)

2. **Heterotropic allosteric:** If the two ligands are different (*e.g.* O₂ BPG),

❖ Macromolecules that have **multiple ligand binding sites** (*e.g.* Hb), allosteric effects can generate cooperative behavior.

❖ Allosteric effects are important in **the regulation of enzymatic reactions**.

❖ Both **allosteric activators (which enhance activity)** and **allosteric inhibitors (which reduce activity)** are utilized to control enzyme reactions.

Allosteric effects

- Allosteric effects require the presence of two forms of the macromolecule.
- ✓ One form, **T or tense state**, binds the primary ligand (*e.g.* oxygen) with low affinity.
- ✓ The other form, **R or relaxed state**, binds ligand with high affinity.
- The T and R states are in equilibrium with each other.
- In the case of positive cooperativity the fraction of T states exceeds that of the R state.

The quaternary structure of **oxy-Hb** is described as **R (relaxed)** form; & **deoxy- Hb** is **T (tight)** form.

'T' form (Taut form) or (Tense form)

1. Deoxygenated Hb
2. The rigid subunit $\alpha_1\beta_1$ and $\alpha_2\beta_2$ are close to the long axis
3. Salt bridges are numerous and plenty and intact
4. Valine residue covers the haem pocket of β -subunit and does not allow entry of O_2
5. BPG can bind and helps in retaining the salt bridges
6. Fe^{++} is 0.07 nm out of the plane of porphyrin ring
7. Affinity of O_2 is low
8. β -chain Histidine residues are protonated (H^+ added)

'R' form (Relaxed form)

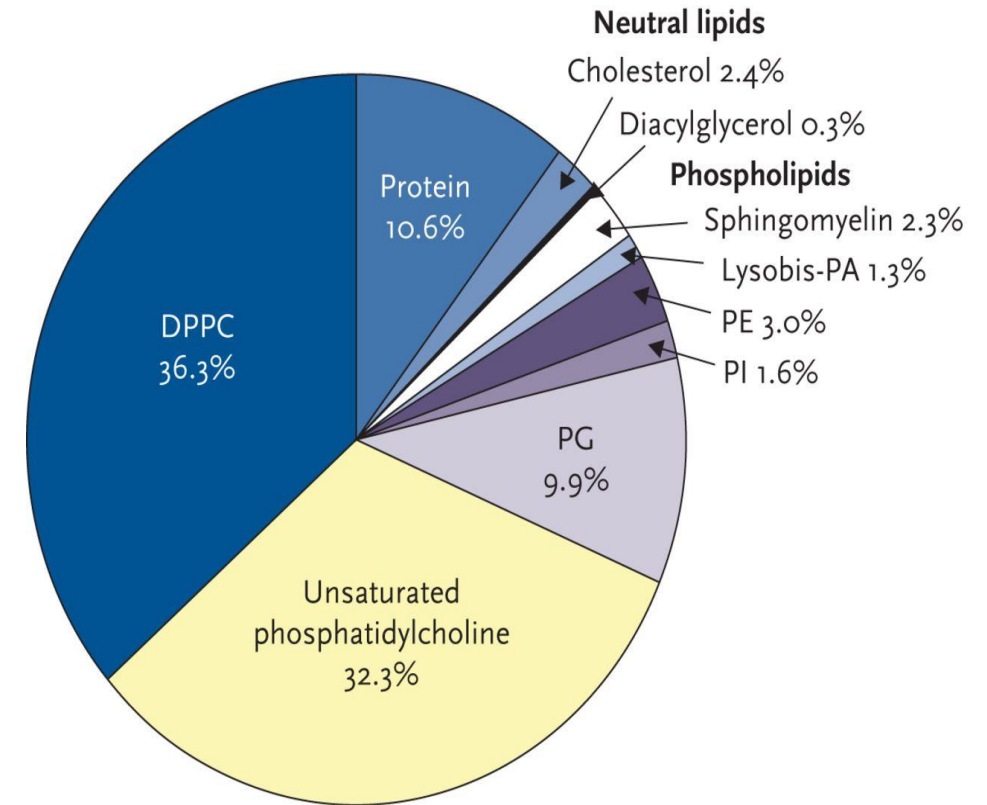
1. Oxygenated Hb
2. Rotation of 15° along the axis
3. Due to rotation the salt bridges are broken
4. Valine residue in haem pocket of β -subunit is removed and O_2 can enter
5. BPG cannot bind as the salt bridges are broken
6. Fe^{++} comes in plane of porphyrin ring
7. Affinity to O_2 becomes high (Haem-haem interaction)
8. Histidines of β -chains release protons (H^+)

Surfactant

- **Definition** :Lung surfactant is a complex mixture of **lipids(90%)** & **proteins(10%)** that lines the alveoli (air sacs) in the lungs.

- **Chemical Composition**

- .DPPC: dipalmitoylphosphatidylcholine
- PA: phosphatidic acid
- PE: phosphatidylethanolamine
- PG: phosphatidylglycerol
- PI: phosphatidylinositol



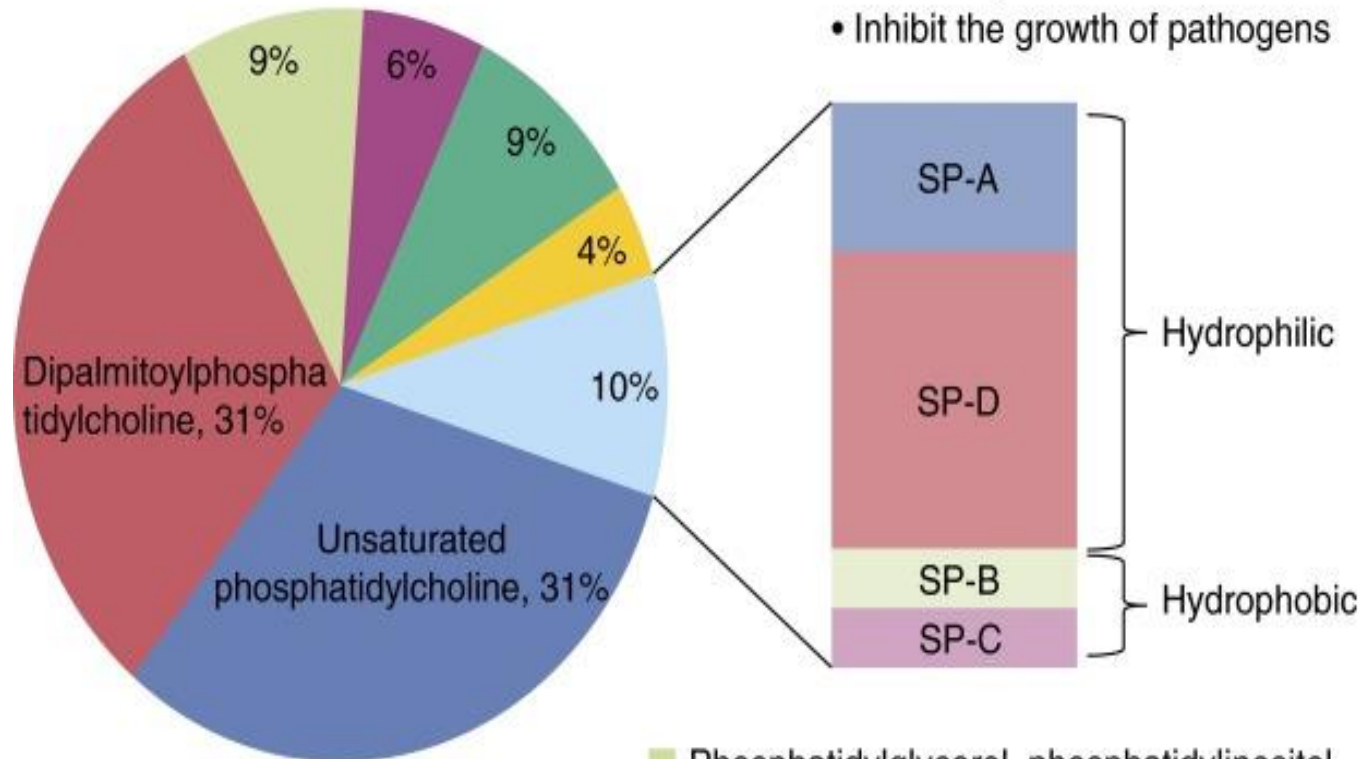
The composition & function of surfactant

Surfactant lipids (90%)

- Lower surface tension
- Change proliferation and cytotoxicity of lymphocytes

Surfactant proteins (10%)

- Enhance chemotaxis and phagocytosis
- Aggregation and opsonization of micro-organisms
- Inhibit the growth of pathogens



Phosphatidylglycerol, phosphatidylinositol

Phosphatidylserine, phosphatidylethanolamine, and sphingomyelin

Neutral lipids

Other lipids

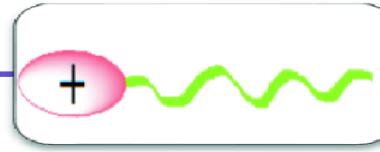
D/t types of Surfactants & application

Classification

Applications

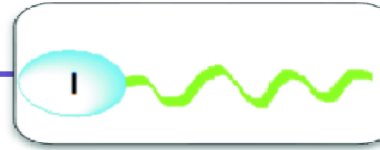
Surfactants

Cationic surfactant



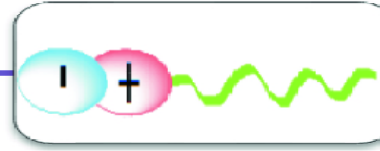
Used to treat burns and wounds due to their excellent disinfectant and bactericidal properties.

Anionic surfactants



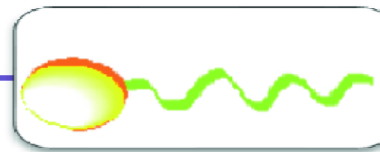
Used in preoperative skin cleaner, medicated shampoos and have bacteriostatic activity.

Zwitterionic
(Amphoteric)
surfactant

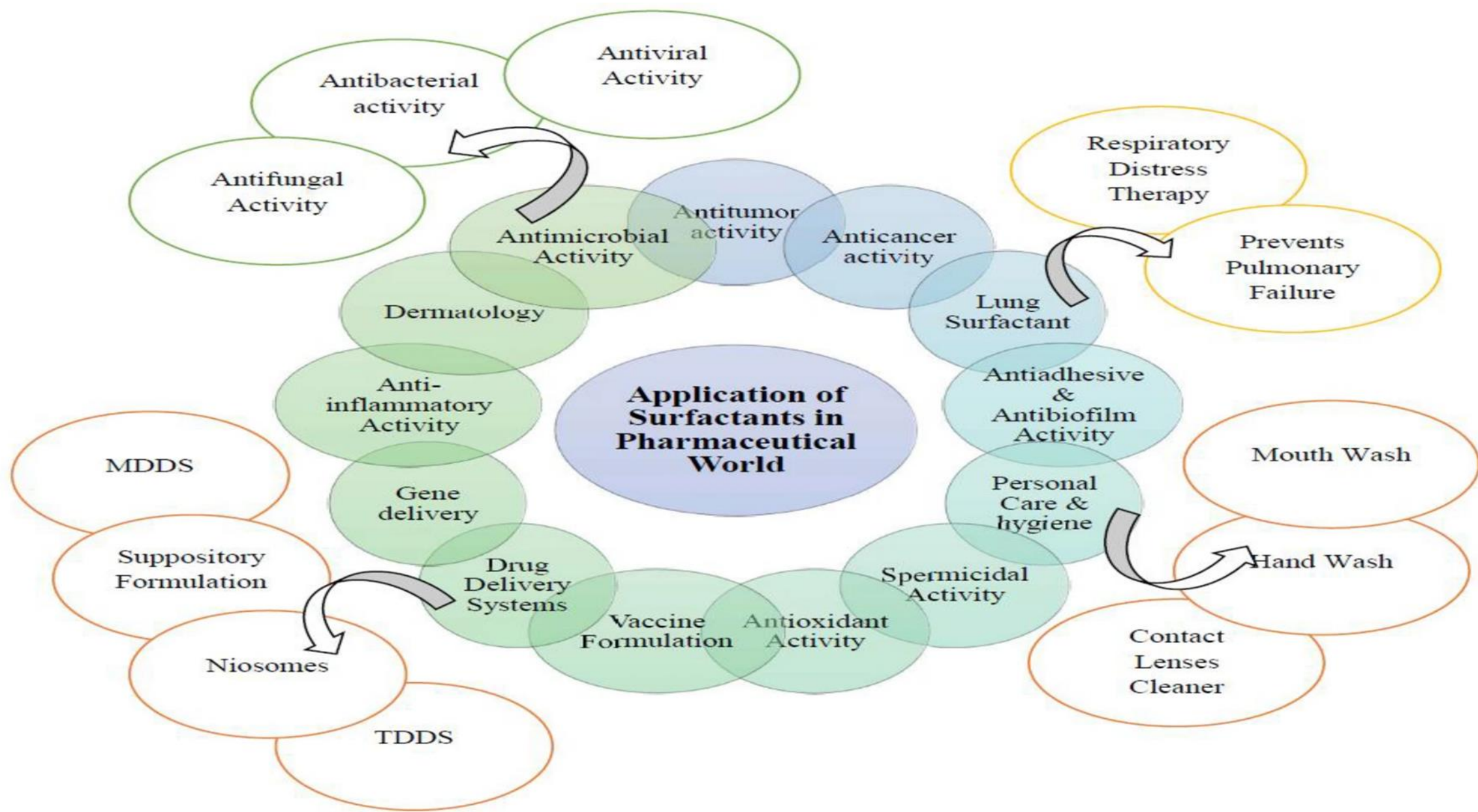


Have excellent dermatological properties.

Non-ionic Surfactant



Act as nano carriers like niosomes, liposomes, and polymersomes.



Clinical Use

- Role in neonatal respiratory distress syndrome (NRDS)
- Surfactant replacement therapy
- Other respiratory diseases (e.g., ARDS)
- RDS is a common breathing disorder that affects newborns
- RDS occurs most often in babies born before their due date,
- usually before 28 weeks of pregnancy.
- Less often, RDS can affect full-term newborns.
- Most newborns who have RDS survive

Symptoms

- Fast & shallow breathing
- Grunting
- Flaring of the nostrils with each breath
- Bluish tone to a baby's skin & lips
- Pulling inward of the muscles between the ribs when breathing

Tests

- **Lung imaging tests**: chest X-rays
- **Heart tests**: an echocardiogram can check for a possible **congenital heart defect**
- **Blood tests** look for an infection